

ER Diagram Notation Comparison

There are three popular notations.

| Chen Notation

Most common in universities and interviews.

Uses:

- Rectangles → Entities
- Diamonds → Relationships
- Ovals → Attributes

| Crow's Foot Notation

Widely used in industry and database design tools.

Uses:

- Boxes for entities.
- Lines with crow's feet for cardinality.
- Attributes listed inside entity boxes.

| ER Design Methodology

Whenever you're given a paragraph in an interview, follow this order.

Step 1: Read the requirements carefully.

Step 2: Identify nouns. Likely entities.

Step 3: Identify verbs. Likely relationships.

Step 4: Identify attributes. Ask: *"What information do we store about each entity?"*

Step 5: Find primary keys.

Step 6: Determine cardinality.

Step 7: Determine participation.

Step 8: Look for:

- Weak Entities
- ISA
- Aggregation
- Recursive Relationships

Step 9: Validate against business rules.

Case Study 1 — College Management System

Identify Attributes

STUDENT

- StudentID (PK)
- RollNumber (Unique)
- Name
- DOB
- Gender
- Phone
- Email
- Address
- AdmissionYear
- Semester
- DepartmentID (FK)

Example: StudentID: 101, Name: Aadya, Department: CSE, Semester: 5

PROFESSOR

- ProfessorID (PK)
- Name
- Phone
- Email
- Qualification
- Salary
- DepartmentID (FK)

DEPARTMENT

- DepartmentID (PK)
- DepartmentName
- Building
- OfficeNumber

COURSE

- CourseID (PK)
- CourseCode
- CourseName
- Credits
- DepartmentID (FK)

ENROLLMENT

- EnrollmentID (PK)
- StudentID (FK)
- CourseID (FK)
- Semester
- Grade

CLASSROOM & EXAM

CLASSROOM

- ClassroomID (PK)
- RoomNumber
- Building
- Capacity

EXAM

- ExamID (PK)
- CourseID (FK)
- ExamType
- Date
- MaximumMarks

ATTENDANCE & BOOK

ATTENDANCE

- AttendanceID (PK)
- StudentID (FK)
- CourseID (FK)
- LectureDate
- Status

BOOK

- BookID (PK)
- ISBN
- Title
- Author
- Publisher
- Category

BOOK ISSUE

Weak/Associative Entity

- IssueID (PK)
- BookID (FK)
- StudentID (FK)
- IssueDate
- ReturnDate

HOSTEL & HOSTEL ROOM

HOSTEL

- HostelID (PK)
- HostelName
- Type
- Capacity

HOSTEL ROOM

- RoomID (PK)
- HostelID (FK)
- RoomNumber
- Floor
- Capacity

FEE PAYMENT

- PaymentID (PK)
- StudentID (FK)
- Amount
- PaymentDate
- Mode
- Status

4. Primary Keys

Entity	Primary Key
Student	StudentID
Professor	ProfessorID
Department	DepartmentID
Course	CourseID
Enrollment	EnrollmentID
Classroom	ClassroomID
Exam	ExamID
Attendance	AttendanceID
Book	BookID
BookIssue	IssueID
Hostel	HostelID
HostelRoom	RoomID
FeePayment	PaymentID

5. Candidate Keys

- **Student:** StudentID, RollNumber
- **Course:** CourseID, CourseCode
- **Book:** BookID, ISBN

6. Relationships

Relationship	Description
Department	Has Students
Department	Employs Professors
Department	Offers Courses
Student	Enrolls in Course
Professor	Teaches Course
Course	Has Exams
Student	Takes Exam
Student	Borrows Book
Hostel	Contains Rooms
Student	Lives In Room
Student	Pays Fee

7. Cardinality

Relationship	Cardinality
Department → Student	1:N
Department → Professor	1:N
Department → Course	1:N
Student → Enrollment	1:N
Course → Enrollment	1:N
Professor → Course	1:N (simplified assumption)
Course → Exam	1:N
Student → Attendance	1:N
Course → Attendance	1:N
Student → BookIssue	1:N
Book → BookIssue	1:N
Hostel → Room	1:N
Room → Student	1:N (assuming shared rooms)
Student → FeePayment	1:N

8. Participation Constraints

Relationship	Participation
Student → Department	Total
Department → Student	Partial
Professor → Department	Total
Department → Professor	Partial
Course → Department	Total
Department → Course	Partial
Enrollment → Student	Total
Enrollment → Course	Total
FeePayment → Student	Total
Student → FeePayment	Partial

9. Weak Entities

BookIssue

Cannot exist without:

- Student
- Book

Therefore, Weak Entity.

HostelRoom can also be modeled as weak if room numbers are unique only within a hostel (e.g., Room 101 in Hostel A and Room 101 in Hostel B).

10. Generalization / Specialization

```
Person
|
└─ Student
```

|
└ Professor

Common attributes: PersonID, Name, DOB, Gender, Phone, Address

Student: RollNumber, Semester

Professor: Salary, Qualification

Inheritance removes duplication.

| 11. Complete ER Diagram

erDiagram

```
DEPARTMENT ||--o{ STUDENT : admits
DEPARTMENT ||--o{ PROFESSOR : employs
DEPARTMENT ||--o{ COURSE : offers
STUDENT ||--o{ ENROLLMENT : enrolls
COURSE ||--o{ ENROLLMENT : contains
PROFESSOR ||--o{ COURSE : teaches
COURSE ||--o{ EXAM : has
STUDENT ||--o{ ATTENDANCE : records
COURSE ||--o{ ATTENDANCE : contains
STUDENT ||--o{ BOOK_ISSUE : borrows
BOOK ||--o{ BOOK_ISSUE : issued
HOSTEL ||--o{ HOSTEL_ROOM : contains
HOSTEL_ROOM ||--o{ STUDENT : accommodates
STUDENT ||--o{ FEE_PAYMENT : pays
DEPARTMENT {
    int DepartmentID PK
    string DepartmentName
    string Building
}
STUDENT {
    int StudentID PK
    string RollNumber
    string Name
    string Email
    int DepartmentID FK
}
PROFESSOR {
    int ProfessorID PK
    string Name
    string Qualification
```

```
        int DepartmentID FK
    }
    COURSE {
        int CourseID PK
        string CourseCode
        string CourseName
        int DepartmentID FK
    }
    ENROLLMENT {
        int EnrollmentID PK
        int StudentID FK
        int CourseID FK
        string Semester
    }
    EXAM {
        int ExamID PK
        int CourseID FK
        string ExamType
    }
    ATTENDANCE {
        int AttendanceID PK
        int StudentID FK
        int CourseID FK
    }
    BOOK {
        int BookID PK
        string ISBN
        string Title
    }
    BOOK_ISSUE {
        int IssueID PK
        int StudentID FK
        int BookID FK
    }
    HOSTEL {
        int HostelID PK
        string HostelName
    }
    HOSTEL_ROOM {
        int RoomID PK
        int HostelID FK
    }
    FEE_PAYMENT {
        int PaymentID PK
```

```
int StudentID FK
decimal Amount
}
```

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